UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*3240199220*

**PHYSICS** **5054/03**

Paper 3 Practical Test **October/November 2009**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
For each of the questions in Section A, you will be allowed to work with the apparatus for a maximum of 20 minutes. For the question in Section B, you will be allowed to work with the apparatus for a maximum of 1 hour.
You are expected to record all your observations as soon as these observations are made.
An account of the method of carrying out the experiments is **not** required.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 | |
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| 3 | |
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| Total | |

This document consists of **9** printed pages and **3** blank pages.


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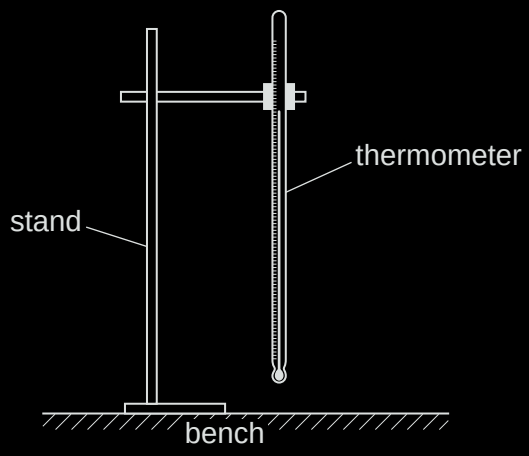



**Fig. 1.1**

Remove the thermometer and quickly support it in the clamp, as shown in Fig. 1.1.

**(b)** When the temperature reaches 60.0°C, start the stopwatch and record the time $t_1$ for the temperature to fall to 50.0°C.

$t_1$ = ..... [1]

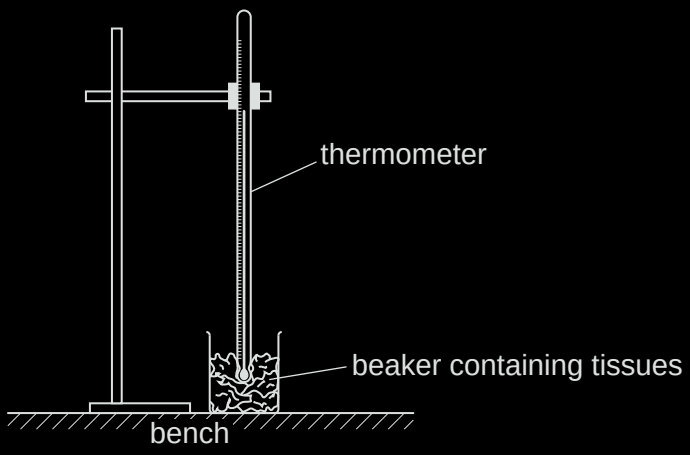



**Fig. 1.2**

The bulb of the thermometer should be placed in the middle of the tissues in the beaker.

**(d)** When the temperature reaches 60.0°C, start the stopwatch and record the time $t_2$ for the temperature to fall to 50.0°C.

$t_2$ = ..... [1]

**(e)** The average rate of fall of temperature is given by

$$\frac{\text{temperature fall}}{\text{time taken for the fall}}.$$

**(i)** Determine the average rate of fall of temperature for both the thermometer in air and the thermometer surrounded by tissues.

rate of fall of temperature in air = .....

rate of fall of temperature in tissues = .....

**(ii)** State the effect that the tissues have had on the rate at which the temperature falls.

.....

..... [2]

**(f)** Suggest why you were instructed to start and stop timing at the same temperature in both **(b)** and **(d)**.

.....

..... [1]

[Total: 5]

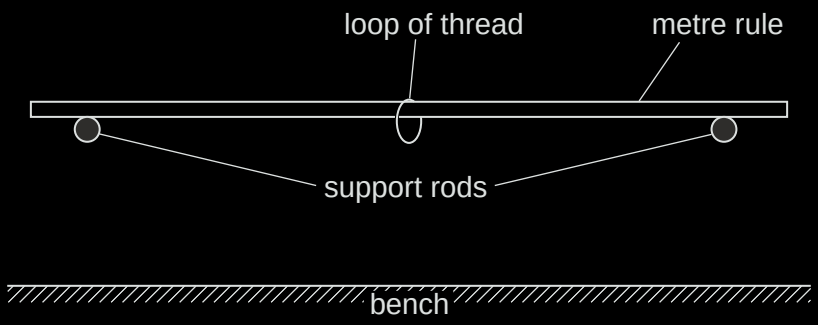



**Fig. 2.1**

The support rods should be at the 5.0 cm and 95.0 cm marks on the metre rule and the loop of thread should be at the 50.0 cm mark.

**(b)** Clamp the half-metre rule vertically, with its scale facing towards you. The zero end of the half-metre rule should be uppermost. The half-metre rule should be placed behind the centre of the metre rule.

**(c)** Suspend a mass $M$ = 0.500 kg from the centre of the metre rule, as shown in Fig. 2.2.

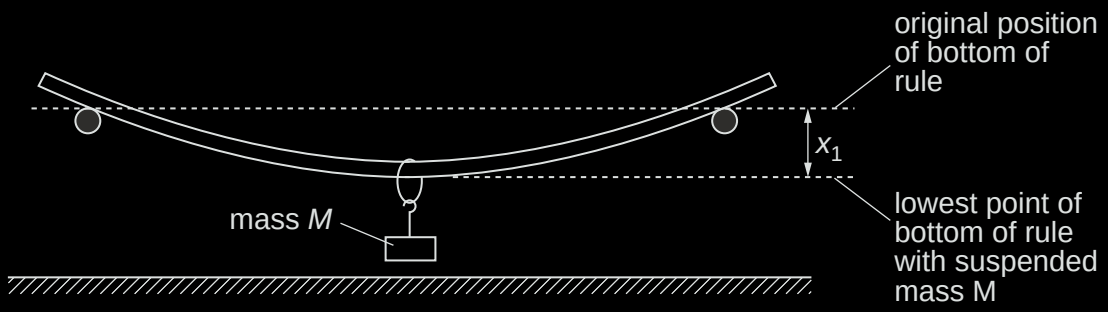



**Fig. 2.2**

Determine the vertical displacement $x_1$ of the centre of the rule, as shown in Fig. 2.2. Show the two readings that you took from the half-metre rule in order to determine $x_1$.

$x_1$ = ..... [2]

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